4.3 Inverse Proportion

Let us consider the following situation.

Situation The following table shows the number of workers and the number of days to finish the construction of water tank in school.

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Number of workers (x)	2	4	5	10
Number of days (y)	40	20	16	8

In this situation, the two quantities namely number of workers and the number of days are related to each other. We observe that when the number of workers increases, the number of days to finish the job decreases. Consider the number of workers as x and the number of days as y then we note that the product is always the same. That is,

$$x \times y = 2 \times 40 = 4 \times 20 = 5 \times 16 = 10 \times 8$$

Consider each of the value of x and the corresponding value of y. Their products are all equal say $xy = 80 = k$ (k is a constant) and it can be expressed as $xy = k$ (k is a constant)

If y_1 and y_2 are the values of y corresponding to the values of x_1 and x_2 of x respectively then $x_1 y_1 = x_2 y_2$ $(=k)$ $\frac{x_1}{y_1} = \frac{x_2}{y_2}$. We say that x and y are in inverse proportion.

Think

Think of an example in real life where two variables are inversely proportional.

Try these

1. Complete the table given below and find the type of proportion

No. of chocolates	1	___	3	4	5	___	____ proportion
Price in rupees (₹)	5	10	___	20	___	___	

No. of workers	1	2	4	5	___	___	___	____ proportion
Time in hours	20	___	5	___	2	1	___	

2. Read the following examples and group them in two categories

S. No.	Quantities	Direct proportion	Inverse proportion
1.	Number of note books purchased and its cost.		
2.	Food shared by number of students to the fixed quantity of food.		
3.	Number of boxes of same size and their weight		
4.	Number of uniforms to the number of students.		
5.	Speed of a vehicle and the time taken to cover the fixed distance		

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